YolApp (Pothole Detection App) - User Interface Guide

Document purpose: This document describes every screen, button, label, message and behavior in the user interface of the mobile app named **YolApp** (repository name: pothole-detection). It is written as reference material for a retrieval (RAG) knowledge base so that a chatbot can answer questions about what the app contains and how to use it. Each section is self-contained and repeats the screen name so that it still makes sense when retrieved alone.

1. App Overview

YolApp is a React Native (version 0.87) mobile app that runs on Android (and iOS). It is built to support pothole detection work. Currently the app has **three screens** and **one floating chatbot**:

- **Home screen** - the starting screen with two navigation buttons.
- **File Upload screen** - lets the user pick a file from the device and view its details (and preview if it is an image).
- **Motion Detection screen** (title inside the screen: "AI Motion Data Recorder") - records accelerometer (motion sensor) data and exports it as a CSV file.
- **Chatbot widget** (assistant named "Hsk", subtitle "Ask about potholes") - a floating chat window available on every screen.

The app has no login, no sign-up, no settings page, no map and no camera screen at present. It works as a data-collection tool: motion data and files can be gathered on the phone, for example to be used later for training or testing an AI model for pothole detection.

2. App Structure and Navigation

The app uses a stack navigator: the Home screen is the first screen (initial route). Pressing a button on Home opens another screen on top. The user returns to Home using the back arrow in the top header bar (or the Android back button).

Screen	Route name	Header title in top bar	How to reach it
Home	Home	Home	Opens when the app starts
File Upload	FileUpload	File Upload	Home -> button "Go to File Upload"
Motion Detection	MotionDetection	Motion Detection	Home -> button "Go to Motion Detection"

The chatbot is not a separate screen. It floats above all screens, in the bottom-right corner, so the user can use it from Home, File Upload and Motion Detection alike.

3. Home Screen

Screen name: Home. **Header title:** "Home". This is the first screen shown when YolApp opens.

3.1 Layout

Content is centered vertically and horizontally on the screen, with a gap between items. From top to bottom: a title text, then two blue buttons.

3.2 Elements on the Home screen

Element	Type	Text shown	What it does
Title	Text (bold, size 20)	Home	Names the screen. Not clickable.

Element	Type	Text shown	What it does
Button 1	Blue rounded button, white text	Go to File Upload	Navigates to the File Upload screen.
Button 2	Blue rounded button, white text	Go to Motion Detection	Navigates to the Motion Detection screen.
Chatbot button	Floating round button (bottom-right)	H followed by a robot emoji	Opens or closes the chatbot (see section 6).

Both buttons use the same blue color (#2f6fed), rounded corners, and white semi-bold text.

4. File Upload Screen

Screen name: File Upload. **Header title:** "File Upload". **Title inside the screen:** "File Upload Page". Reached from Home by pressing "Go to File Upload".

4.1 Purpose

The File Upload screen lets the user choose one file from the phone's storage using the system file picker. The app then shows the file's name, type and size. If the chosen file is an image, the user can also display a preview. Note: the screen only *selects* and displays the file; the code does not send it to a server.

4.2 Elements on the File Upload screen

Element	Type	Text shown	What it does
Title	Text (bold, size 20)	File Upload Page	Names the screen.
Upload button	Blue rounded button	Upload File	Opens the device's file picker. Any file type is allowed (all files). Only one file can be chosen at a time (multi-select is off).
Selected file card	Light-blue card with border (88% width)	Appears only after a file is picked	Shows information about the chosen file (see 4.3).

4.3 The Selected File card

The card is hidden until the user picks a file. It contains, from top to bottom:

- **Label "SELECTED FILE"** - small uppercase grey heading (written "Selected file" in code, displayed in uppercase).
- **Image preview** - a 180-pixel-high picture of the file. Shown **only if** the file is an image (its type starts with "image/") **and** the user has pressed "Show Preview". Hidden by default.
- **File name** - bold dark text, up to two lines. If the file has no name, "Unnamed file" is shown.
- **File details line** - the file type (for example image/jpeg or application/pdf) or "Unknown file type" if not known, followed by " | " and the size if the size is known.
- **"Remove file" link** - red text. Clears the selection and hides the card.
- **"Show Preview" link** - blue text. Toggles the image preview on and off. The wording stays "Show Preview" in both states. For non-image files (PDF, CSV etc.) pressing it does nothing visible because only images can be previewed.

4.4 How file size is displayed

- Under 1 KB (less than 1024 bytes): shown in bytes, for example "512 B".

- 1024 bytes or more: shown in megabytes with one decimal, for example "2.4 MB". (Small files between 1 KB and 100 KB therefore appear as "0.0 MB" or "0.1 MB"; there is no KB unit.)

4.5 Messages and error handling

- If the user closes the picker without choosing a file (cancel), nothing happens and no message appears.
- If the picker fails for any other reason, an alert pops up with the title **"Unable to select file"** and the message **"Please try again."**

4.6 Step-by-step: how to upload and preview an image

- 1. On Home, tap "Go to File Upload".
- 2. Tap "Upload File".
- 3. In the system picker, choose a file.
- 4. The Selected File card appears with the name, type and size.
- 5. If it is an image, tap "Show Preview" to display it; tap again to hide it.
- 6. Tap "Remove file" to clear the selection and start over.

5. Motion Detection Screen (AI Motion Data Recorder)

Screen name: Motion Detection. **Header title:** "Motion Detection". **Title inside the screen:** "AI Motion Data Recorder". Reached from Home by pressing "Go to Motion Detection". The screen has a light grey background.

5.1 Purpose

This screen records the phone's **accelerometer** (motion sensor) values along the X, Y and Z axes while the user drives or moves with the phone, then exports the recording as a **CSV table** that can be opened in a spreadsheet app such as Excel or Google Sheets. The recorded data is intended for AI model training or analysis (for example, detecting potholes from vibration patterns). The screen does not itself detect potholes; it only records and exports data.

5.2 Elements on the Motion Detection screen

Element	Type	Text shown	What it does
Title	Text (bold, size 20)	AI Motion Data Recorder	Names the screen.
Row counter	Text	Rows Collected: **N**	Shows how many sensor readings have been recorded in the current session. N starts at 0 and rises while recording. The two asterisks around the number are literally displayed on screen (Markdown bold is not rendered).
Start button	Green button (#22c55e)	Start Recording	Shown when not recording. Clears any previous recording, resets the counter to 0 and starts collecting accelerometer data.
Stop button	Red button (#ef4444)	Stop Recording	Replaces the Start button while recording. Stops collecting data; the collected rows are kept in memory.

Element	Type	Text shown	What it does
Export button	Default (blue) button	Export Table (CSV)	Saves the recorded rows as a CSV file. Disabled (greyed out) while recording is in progress.

The Start and Stop buttons are the same button slot: only one of them is visible at a time.

5.3 Recording details

- **Sensor:** accelerometer (x, y, z acceleration).
- **Sampling rate:** one reading every 20 milliseconds, which is 50 readings per second (50 Hz). This fixed rate is chosen because AI models need a steady, predictable sampling rate.
- **Each row contains:** timestamp, x, y, z.
- Starting a new recording **erases** the previous unsaved recording.
- Leaving the screen automatically stops recording (cleanup on unmount).
- Data is held in memory only until exported; it is lost if the app is closed before exporting.

5.4 CSV export format

The exported file is plain text (UTF-8) with a header line followed by one line per reading:

```
timestamp,accel_x,accel_y,accel_z
1730800000000,0.1234,9.7812,0.0456
1730800000020,0.1301,9.7790,0.0471
```

- Columns: **timestamp, accel_x, accel_y, accel_z**.
- x, y and z values are rounded to 4 decimal places.
- File name pattern: **accelerometer_data_<current time in milliseconds>.csv**, for example accelerometer_data_1730800000000.csv.
- MIME type: text/csv.

5.5 What happens when Export Table (CSV) is pressed

- The app writes the CSV into its own document folder first.
- Then the system "save file" dialog opens so the user chooses where to store it (Downloads, Google Drive, Files, etc.).
- On success an alert appears: **"Export Successful"** - Saved "<file name>" to your device. Open it with any spreadsheet app.
- If the user cancels the save dialog, nothing is shown.

5.6 Messages and alerts on this screen

Situation	Alert title	Alert message
Export pressed but nothing recorded (0 rows)	No Data	Record some data before exporting.
Accelerometer missing or unavailable (e.g., on an emulator)	Sensor Error	Accelerometer is not available on this device/emulator. Recording is stopped automatically.
Saving the file failed	Export Failed	The error text returned by the system.

Situation	Alert title	Alert message
File saved	Export Successful	Saved "<file name>" to your device. Open it with any spreadsheet app.

5.7 Step-by-step: how to record and export motion data

- 1. On Home, tap "Go to Motion Detection".
- 2. Tap the green "Start Recording" button. The counter "Rows Collected" starts increasing (about 50 per second).
- 3. Move or drive with the phone.
- 4. Tap the red "Stop Recording" button.
- 5. Tap "Export Table (CSV)" and choose where to save the file.
- 6. Open the saved CSV in a spreadsheet app.

6. Chatbot Widget (assistant "Hsk")

Component name: ChatBot. **Assistant display name:** "Hsk". **Subtitle:** "Ask about potholes". The chatbot is a floating widget available on **every screen** (Home, File Upload and Motion Detection).

6.1 Position and open/closed state

- It sits in the bottom-right corner of the screen (25 px from the right edge, 40 px from the bottom).
- **Floating button:** a dark-blue pill-shaped button labeled "H" with a robot emoji. It is always visible. Tapping it opens the chat window if closed, or closes it if open.
- **The chat window is open by default** when the app starts.
- When the on-screen keyboard appears, the whole chat widget moves up by the keyboard height so the input box is not hidden.

6.2 Parts of the chat window

The chat window is a white rounded panel, 320 px wide and up to 420 px tall, shown above the floating button.

Part	Description
Header	Light-blue bar with the name "Hsk" (bold blue), the subtitle "Ask about potholes", and a round close button marked with the multiplication sign (x) at the right.
Close button	Small round button in the header. Closes the chat window (the floating "H" button stays so it can be reopened).
Message area	Scrollable list of chat bubbles on a very light blue background (up to 260 px tall).
User bubble	Blue bubble with white text, aligned to the right. Contains what the user typed.
Bot bubble	Grey bubble with dark text, aligned to the left. Contains the assistant's reply.
Input box	Text field with the placeholder "Type your message". Supports multiple lines (grows up to 90 px).
Send button	Blue button labeled "Send" next to the input box.

6.3 Greeting message

When the app starts the chat already shows one bot message: **"Hi! I can help you track pothole issues, inspection notes, and safety updates."**

6.4 How sending a message works

- The user types in the input box and presses "Send".
- Empty or whitespace-only messages are ignored. While a previous message is still waiting for a reply, new sends are ignored.
- The user's message appears as a blue bubble and the input box is cleared.
- The app sends the text to the chat server with an HTTP POST request to **https://python-server-8iwc.onrender.com/chat**. The request body is JSON: {"message": "<user text>"}.
- The server answers with JSON containing a field called **reply**; that text is shown as a grey bot bubble.
- If the server cannot be reached or returns an error status, the bot shows: **"Sorry, I couldn't reach the chat service. Please try again."**

Limitations: the conversation history is kept only while the app is running (it is not saved). There is no typing/loading indicator while waiting. The server is hosted on a free-tier service, so the first reply after a period of inactivity may be slow.

7. Visual Style Summary

- Primary color: blue. Navigation buttons #2f6fed; chatbot accent colors #1d4ed8 and #2563eb.
- Recording colors: green #22c55e for Start, red #ef4444 for Stop; red text for "Remove file".
- Screens use plain light backgrounds, centered content, rounded corners (8-18 px) and semi-bold text.
- Top header bar is the default native stack header showing the screen title and a back arrow.
- The app icon was customised (adaptive launcher icon with foreground, background and monochrome layers on Android).

8. Technology Summary

Item	Detail
App name	YolApp (project folder: pothole-detection)
Framework	React Native 0.87 with React 19 and TypeScript
Navigation	React Navigation native stack (@react-navigation/native-stack)
File picking and saving	@react-native-documents/picker (pick and saveDocuments)
Motion sensor	react-native-sensors (accelerometer, RxJS streams)
File writing	react-native-fs
Chat backend	HTTP POST to https://python-server-8iwc.onrender.com/chat
Source files	src/App.tsx, src/navigation/AppNavigator.tsx, src/Home/Home.tsx, src/FileUpload/FileUpload.tsx, src/MotionSensor/MotionDetection.tsx, src/ChatBot/ChatBot.tsx

9. Frequently Asked Questions (FAQ)

Q: What screens does YolApp have?

A: Three screens: Home, File Upload and Motion Detection (titled "AI Motion Data Recorder" inside), plus a floating chatbot named Hsk available on all screens.

Q: What is on the Home screen?

A: A title "Home" and two blue buttons: "Go to File Upload" and "Go to Motion Detection", plus the floating chatbot button.

Q: How do I go back to the Home screen?

A: Use the back arrow in the top header bar or the Android back button.

Q: How do I upload a file in YolApp?

A: Open File Upload from Home, tap "Upload File" and choose a file in the system picker. Any file type is accepted, one file at a time.

Q: Can I preview the file I selected?

A: Only images can be previewed. After selecting an image tap "Show Preview" to show or hide it. Other file types show only name, type and size.

Q: How do I remove the selected file?

A: Tap the red "Remove file" link on the Selected File card.

Q: What information is shown about the selected file?

A: File name, file type (MIME type) and size (in B or MB).

Q: Does the File Upload screen send my file to a server?

A: No. In the current version it only selects the file and shows its details on the device.

Q: What does the Motion Detection screen do?

A: It records the phone's accelerometer (x, y, z) at 50 readings per second and exports the data as a CSV file for AI training or analysis.

Q: How do I record motion data?

A: Tap "Start Recording", move the phone, then tap "Stop Recording". The "Rows Collected" counter shows how many readings were captured.

Q: How do I export motion data?

A: After stopping, tap "Export Table (CSV)" and pick a save location. The file is named accelerometer_data_<timestamp>.csv and has columns timestamp, accel_x, accel_y, accel_z.

Q: Why is the Export button greyed out?

A: It is disabled while recording. Press "Stop Recording" first.

Q: Why do I see 'No Data' when exporting?

A: Nothing has been recorded yet. Record some data first, then export.

Q: Why do I see 'Sensor Error'?

A: The device or emulator has no accelerometer available. Use a real phone with a motion sensor.

Q: Why does the counter show asterisks like **120?**

A: The label "Rows Collected: **N**" displays the asterisks literally; they are only formatting characters and not part of the number.

Q: Is the recorded data saved automatically?

A: No. It exists only in memory until you export it, and is erased when a new recording starts or the app closes.

Q: Who is Hsk?

A: Hsk is the name of the in-app chatbot assistant. Its subtitle is "Ask about potholes".

Q: How do I open or close the chatbot?

A: Tap the floating "H" robot button at the bottom-right to toggle it, or tap the x button in the chat header to close it. It is open by default.

Q: What can I ask the chatbot?

A: Its greeting says it can help track pothole issues, inspection notes and safety updates. Answers come from an online server.

Q: Why does the chatbot say 'Sorry, I couldn't reach the chat service'?

A: The device is offline or the chat server returned an error or is waking up. Check the internet connection and try again.

Q: Is the chat history saved?

A: No. Messages are kept only while the app is open.

Q: Does the app detect potholes automatically?

A: Not yet in the UI. The app currently collects motion data and files and offers a chatbot; there is no detection result screen, camera or map.